



Timothy R. McNealy

CONTACT

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EDUCATION

Bachelor of Science, Mechanical Engineering,
Minor Mathematics
Texas Tech University, Lubbock, TX
1/2001 - 5/2004

INNOVATIONS

History Matching Multi-Porosity Solutions
Patent# US10344591B2
<https://patents.google.com/patent/US10344591B2>

SKILLS

Programming

C# | C++ | Cuda | C | VBA | R | SQL | Matlab |
Python | HTML | JavaScript | Data Analysis |
Mining | PostgreSQL | Apache Thrift | Spotfire |
R Studio | Machine Learning | Neural Networks

CAD

FreeCAD | Blender | SolidWorks

Software

Windows 11 | Ubuntu | Excel | Word |
PowerPoint | Access etc.

Automotive

Import/Domestic Engine Cylinder Head
Machining | Full Engine Disassembly &
Assembly

Oil & Gas

Hydraulic Fracturing | Geology | Petrophysics |
Cement | Coiled Tubing | WAG Flooding |
Conformance | Reservoir Simulation |
Production & Economic Optimization | Gohfer
| FracPro PT | PredictK | Materials Library | IFS
| HalWin | Geolog | Prizm | Landmark |
Petrophysics | Fekete R.T.A. & Well Test |
Landmark Nexus | Quiklook | Landmark
DecisionSpace | TerraVista | PDAT

AWARDS

Eagle Scout, Troop 175, Midland, TX, 1998
https://www.westtexascoutinghistory.net/award_eagle_btc.html

PROFESSIONAL SUMMARY

International Experienced Mechanical Engineer with +25 years programming C/C++/C#, +19 years in upstream Oil & Gas (seven countries), +10 years in CAD & 3D Printing & +2 years Automotive Engine machining, disassembly & assembly & +28 years Automotive vehicle disassembly & assembly.

EXPERIENCE

Lead Engineer & Developer Nindra, Houston, TX	4/2019 - Current
- Developed a High Performance Nonlinear Constrained Global Minima solver & a Laplace transform utilizing the GPU(Kokkos/Cuda). The new solver reduces the time to solve from 8 hours (in Matlab) to 1 minute in C++/C#. - Built a PostgreSQL/Apache Thrift database containing +40 GB of Texas RRC data sources.	
Global Senior Technical Advisor Halliburton, Houston, TX	11/2017 - 4/2019
Built custom Spotfire plugin for importing MicroSeismic data to constantly determine the fracture extents.	
North America Technology Advisor Halliburton, Houston, TX	7/2015 - 11/2017
Built custom flow regime production decline analysis to analyze 360K+ wells in order to determine the economic viability of re-fracturing candidates.	
Technical Lead, Cairn India Project Halliburton, Gurgaon, India	11/2014 - 7/2015
- The first well we designed out produced the entire field by increasing liquid production from 100 bpd to +1,200 bpd. - Production integrated into an economic optimization analysis using Gohfer, Quiklook & DMS.	
Eurasia Technical Team Manager Halliburton, Moscow, Russia	7/2014 - 11/2014
Directly generated revenue while working with Halliburton Consulting on Unconventional Petrophysical analysis & modeling Unconventional production in Nexus.	
Russia Technical Team Manager Halliburton, Moscow, Russia	12/2013 - 7/2014
Engaging & transferring Unconventional knowledge to customers & to all of Halliburton in Russia.	
Completion & Production Applied Engineering Manager Halliburton, Houston, TX	5/2013 - 12/2013
We pioneered Halliburton's Cypher Service, which increased production & lowered costs for operators all over the world, promoted and generated millions in revenue.	
Technical Sales Advisor, South Texas Halliburton, Houston, TX	9/2010 - 5/2013
Halliburton saved millions after I designed & implemented an operational efficiency matrix (stimulation treatments) to track Non-Productive Time (NPT).	
Principal Technical Services Specialist, PB Halliburton, Midland, TX	9/2007 - 9/2010
Optimized horizontal fracture spacing for Yates Petroleum in the 3rd Bone Spring, today it is still one of the highest producing wells per foot in the SENM Delaware Basin.	
Technical Professional Halliburton, Artesia, NM	2/2006 - 9/2007
Sold \$4M of Expedite to Yates Petroleum.	
Associate Technical Professional Halliburton, Alice, TX - Hobbs, NM	1/2005 - 2/2006
Engine Cylinder Head Machinist Robert's Machine Shop, Midland, TX	6/1999 - 12/2000

PUBLICATIONS

Effective Fracture Placement in Horizontal Wells https://www.swpshortcourse.org/papers?combine=Effective+Fracture+Placement+in+Horizontal+Wells	Southwestern Petroleum
Highly Accurate Prediction of Screenouts in the Eagle Ford Shale with the Screenout Index https://www.onepetro.org/conference-paper/SPE-157613-MS	SPE
Proppant economics in the Eagle Ford formation https://www.hartenergy.com/exclusives/proppant-economics-eagle-ford-formation-19236	Hart Energy
Multi-Porosity Modeling for Unconventional Reservoirs https://www.swpshortcourse.org/papers?combine=Multi-Porosity+Modeling+for+Unconventional+Reservoirs	Southwestern Petroleum